Year 10 Mathematics

Shape and Space 1

5.4 Perimeter and Area of Triangles and Quadrilaterals

Perimeter and Area of Triangles and Quadrilaterals

STARTER · 5 MIN

Date: _____ **Title:** Perimeter and Area of Triangles and Quadrilaterals

LEARNING OBJECTIVES

- Find the area of triangles and parallelograms
- Find the area of a trapezium
- Solve perimeter and area problems

STARTER *Answer in your book. No calculators.*

1. Work out 8×6
2. Work out $\frac{1}{2} \times 10 \times 7$
3. Work out $(5 + 9) \div 2$
4. Work out $2 \times (7 + 4)$

Starter — Answers

STARTER

Tick or correct in green pen

1. $8 \times 6 = \mathbf{48}$
2. $\frac{1}{2} \times 70 = \mathbf{35}$
3. $14 \div 2 = \mathbf{7}$
4. $2 \times 11 = \mathbf{22}$

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the area of triangles and parallelograms	☐
SILVER	I can find the area of a trapezium	☐
GOLD	I can solve perimeter and area problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Area

INTRODUCTION · 3 MIN

Area is measured in square units, such as cm^2.

Rectangle

Length × width

$A = l \times w$

Triangle

Half of base × height

$$A = \frac{1}{2} \times b \times h$$

Parallelogram

Base × perpendicular height

$A = b \times h$

Trapezium

Half the sum of the parallel sides × height

$$A = \frac{1}{2}(a + b)h$$

Examples 1 and 2 — Triangles and Parallelograms

EXAMPLES · 5 MIN

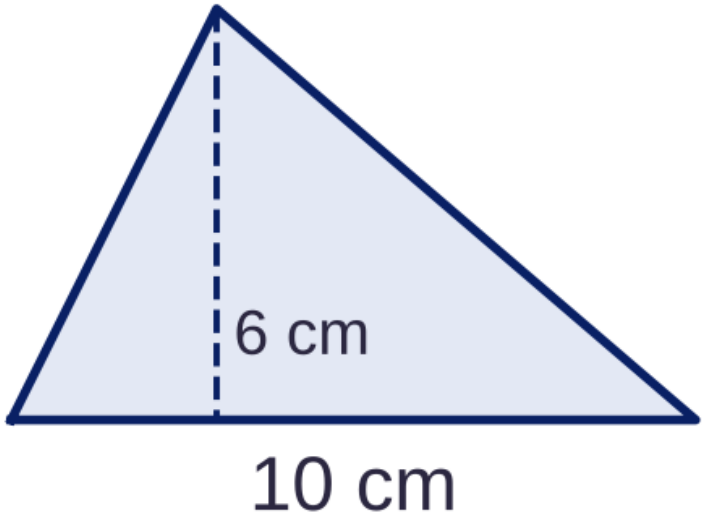
BRONZE

Watch and copy into your book

Hint triangle: $A = \frac{1}{2}bh$ parallelogram: $A = bh$

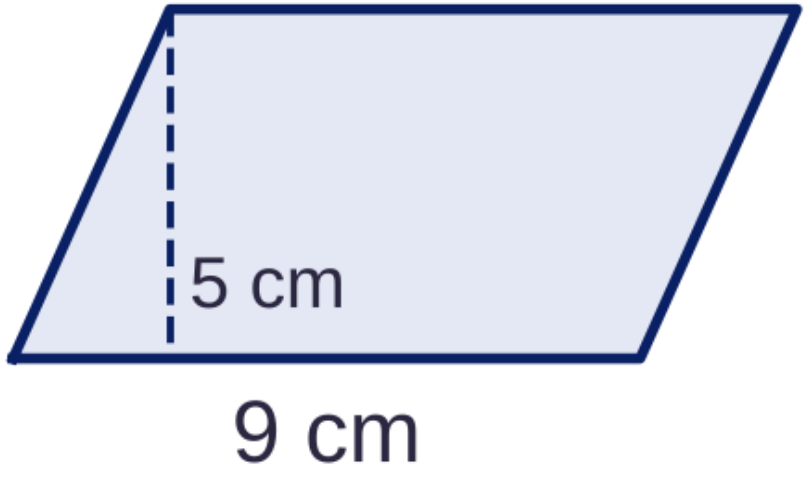
Example 1

Find the area of the triangle.



Example 2

Find the area of the parallelogram.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

$$A = \frac{1}{2} \times 10 \times 6$$

$A = 30\ cm^2$

Tip: use the perpendicular height, not the slant side.

Example 2

$$A = 9 \times 5$$

$A = 45\ cm^2$

Tip: area = base × perpendicular height.

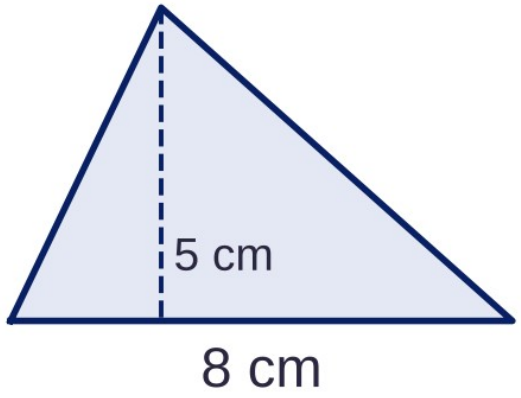
Task 1

PRACTICE · 9 MIN

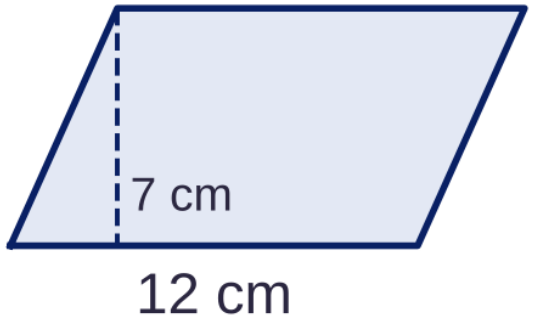
BRONZE *Find the area of each shape*

Hint triangle: $A = \frac{1}{2}bh$ parallelogram: $A = bh$

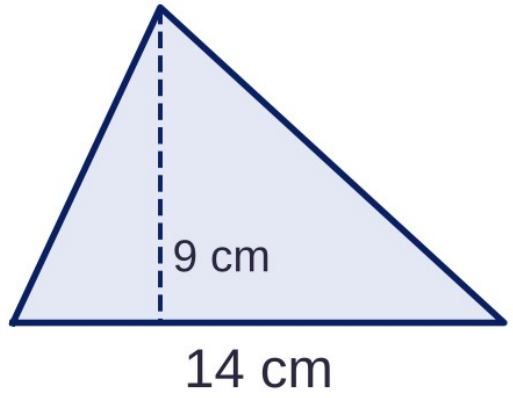
1



2



3



4



5



★ A square has a perimeter of 36 cm. Find its area.

Task 1 — Answers

CORRECTIONS

BRONZE *Tick or correct in green pen*

1. $A = \frac{1}{2} \times 8 \times 5 = \mathbf{20\ cm^2}$

2. $A = 12 \times 7 = \mathbf{84\ cm^2}$

3. $A = \frac{1}{2} \times 14 \times 9 = \mathbf{63\ cm^2}$

4. $A = 6.5 \times 4 = \mathbf{26\ cm^2}$

5. $A = 11 \times 6 = \mathbf{66\ cm^2}$

★ Side $= 36 \div 4 = 9$, $A = 9 \times 9 = \mathbf{81\ cm^2}$

Examples 3 and 4 — Area of a Trapezium

EXAMPLES · 5 MIN

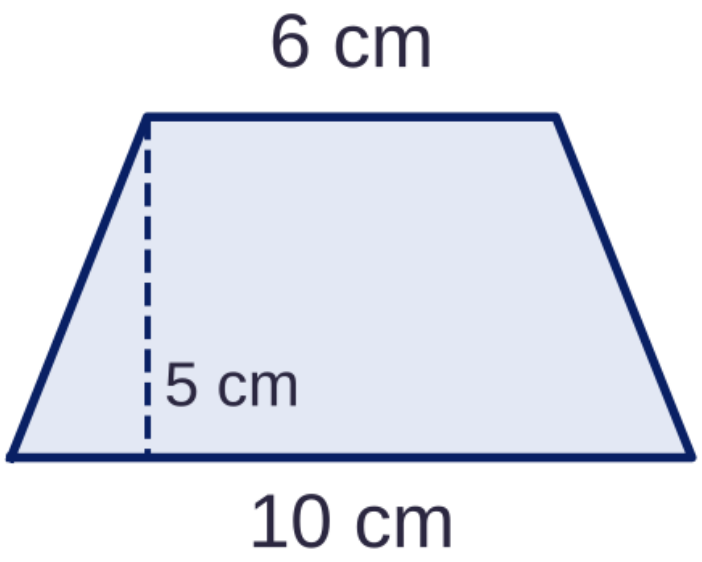
SILVER

Watch and copy into your book

Hint trapezium: $A = \frac{1}{2}(a + b)h$

Example 3

Find the area of the trapezium.



Example 4

A trapezium has parallel sides 4.5 cm and 7.5 cm.

Its height is 4 cm.

Find its area.

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

$$A = \frac{1}{2} \times (6 + 10) \times 5$$

$$A = \frac{1}{2} \times 16 \times 5$$

$A = 40\ cm^2$

Tip: add the parallel sides first.

Example 4

$$A = \frac{1}{2} \times (4.5 + 7.5) \times 4$$

$$A = \frac{1}{2} \times 12 \times 4$$

$A = 24\ cm^2$

Tip: the height is between the parallel sides.

Task 2

PRACTICE · 9 MIN

SILVER

Find the area of each trapezium (a and b are the parallel sides)

Hint trapezium: $A = \frac{1}{2}(a + b)h$

1. a = 5 cm, b = 9 cm, h = 4 cm
2. a = 7 cm, b = 13 cm, h = 6 cm
3. a = 3 cm, b = 11 cm, h = 8 cm
4. a = 2.5 m, b = 6.5 m, h = 5 m
5. a = 10 cm, b = 16 cm, h = 7 cm

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $A = \frac{1}{2} \times 14 \times 4 = \mathbf{28\ cm^2}$

2. $A = \frac{1}{2} \times 20 \times 6 = \mathbf{60\ cm^2}$

3. $A = \frac{1}{2} \times 14 \times 8 = \mathbf{56\ cm^2}$

4. $A = \frac{1}{2} \times 9 \times 5 = \mathbf{22.5\ m^2}$

5. $A = \frac{1}{2} \times 26 \times 7 = \mathbf{91\ cm^2}$

Examples 5 and 6 — Working Backwards

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint triangle: $A = \frac{1}{2}bh$ trapezium: $A = \frac{1}{2}(a + b)h$

Example 5

(2 marks)

A triangle has an area of 42 cm^2.

Its base is 12 cm.

Find its height.

Example 6

(3 marks)

A trapezium has area 60 cm^2.

Its parallel sides are 7 cm and 13 cm.

Find its height.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

$\frac{1}{2} \times 12 \times h = 42$

$6h = 42$

$h = 7$ cm

Tip: write the formula, then solve.

Example 6

$\frac{1}{2} \times (7 + 13) \times h = 60$

$10h = 60$

$h = 6$ cm

Tip: simplify $\frac{1}{2} \times 20$ first.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint triangle: $A = \frac{1}{2}bh$ trapezium: $A = \frac{1}{2}(a + b)h$

1. A parallelogram has area 72 cm² and base 9 cm. Find its height. (2 marks)

2. A triangle has area 54 cm² and height 9 cm. Find its base. (2 marks)

3. A trapezium has parallel sides 6 cm and 10 cm and area 48 cm². Find its height. (3 marks)

4. A rectangle has perimeter 30 cm and length 9 cm. Find its area. (3 marks)

5. A square has area 49 cm². Find its perimeter. (2 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $9h = 72, \quad h = \mathbf{8\ cm}$

2. $\frac{1}{2} \times b \times 9 = 54, \quad 4.5b = 54, \quad b = \mathbf{12\ cm}$

3. $\frac{1}{2} \times 16 \times h = 48, \quad 8h = 48, \quad h = \mathbf{6\ cm}$

4. $\text{Width} = (30 - 18) \div 2 = 6, \quad A = 9 \times 6 = \mathbf{54\ cm^2}$

5. $\text{Side} = \sqrt{49} = 7, \quad P = 4 \times 7 = \mathbf{28\ cm}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Find the area of a triangle with base 12 cm and height 5 cm.

SILVER Find the area of a trapezium with parallel sides 4 cm and 8 cm and height 5 cm.

GOLD A triangle has area 36 cm^2 and base 8 cm. Find its height.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $A = \frac{1}{2} \times 12 \times 5 = \mathbf{30\ cm^2}$

SILVER $A = \frac{1}{2} \times 12 \times 5 = \mathbf{30\ cm^2}$

GOLD $4h = 36, \quad h = \mathbf{9\ cm}$

Exam Practice — Compound Shapes

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint triangle: $A = \frac{1}{2}bh$ parallelogram: $A = bh$

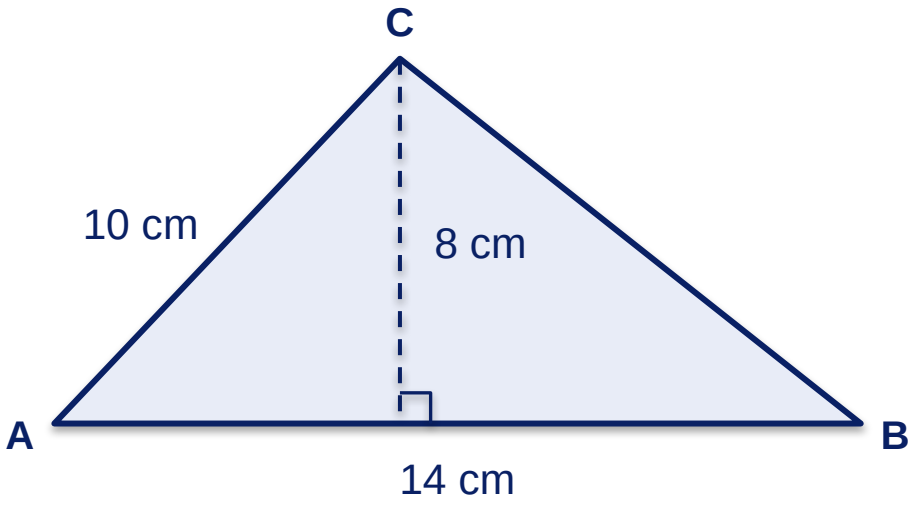
Question 1

(2 marks)

The diagram shows triangle ABC.

Work out the area of the triangle.

Diagram NOT accurately drawn



..... cm²

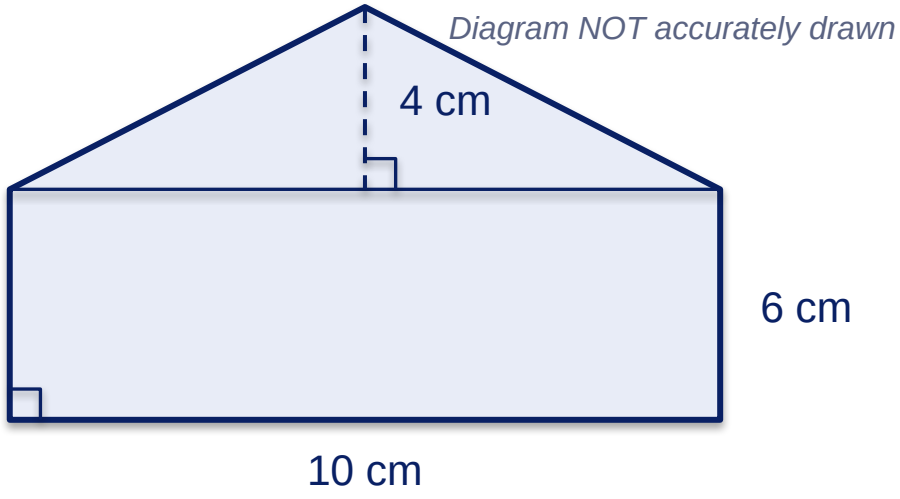
Question 2

(3 marks)

The shape is made from a rectangle and a triangle.

Work out the total area.

Diagram NOT accurately drawn



..... cm²

Exam Practice — Trapeziums in Context

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint trapezium: $A = \frac{1}{2}(a + b)h$

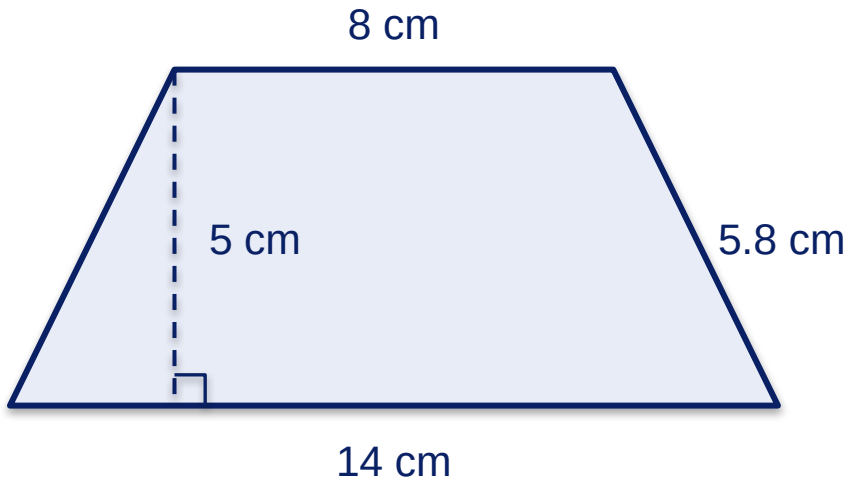
Question 3

(2 marks)

The diagram shows a trapezium.

Work out the area of the trapezium.

Diagram NOT accurately drawn



..... cm^2

Question 4

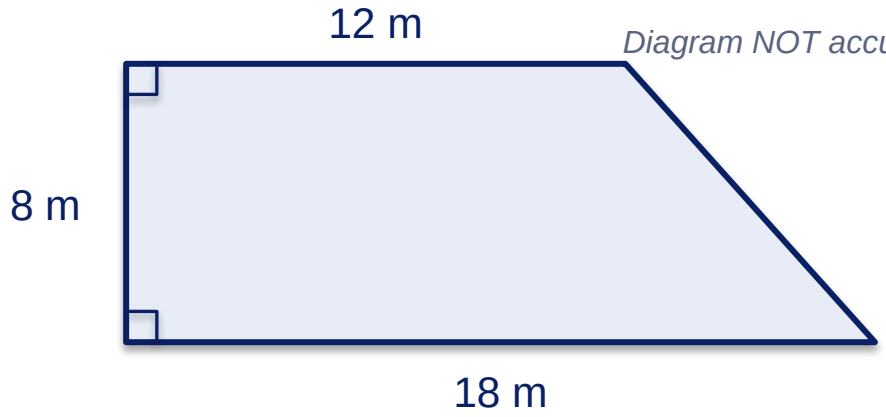
(3 marks)

Mr Ali's lawn is a trapezium.

One box of grass seed covers 25 m^2.

How many boxes must he buy to cover the lawn?

Diagram NOT accurately drawn



..... boxes

Exam Practice — Forming Equations

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint triangle: $A = \frac{1}{2}bh$ trapezium: $A = \frac{1}{2}(a + b)h$

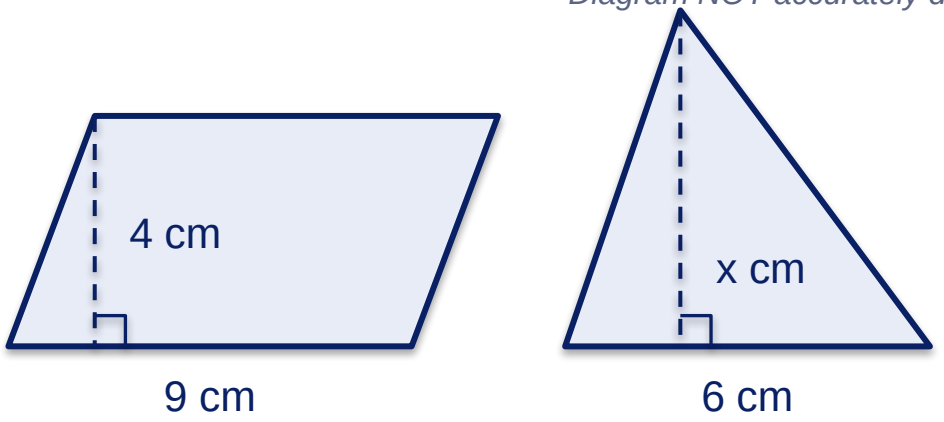
Question 5

(3 marks)

The parallelogram and the triangle have the same area.

Work out the value of x.

Diagram NOT accurately drawn



x = ...

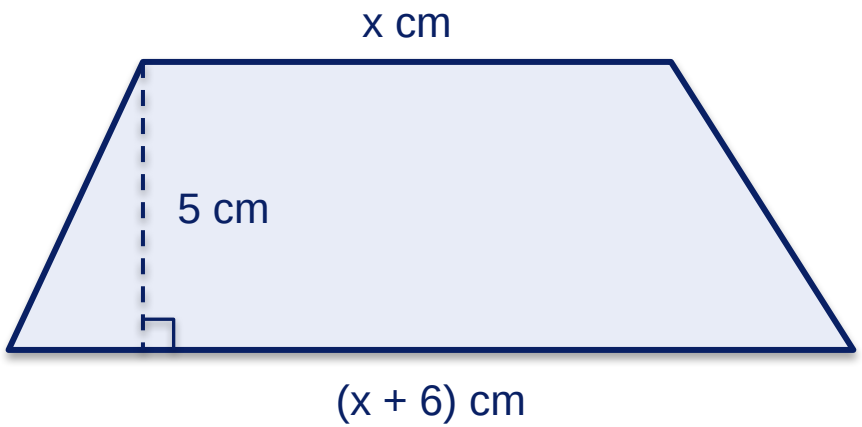
Question 6

(4 marks)

The area of the trapezium is 65 cm^2.

Work out the value of x.

Diagram NOT accurately drawn



x = ...

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

$\frac{1}{2} \times 14 \times 8$ — M1

= 56 cm² — A1

Tip: 10 cm is the slant side, so don't use it.

Question 2

$10 \times 6 = 60$ — M1

$\frac{1}{2} \times 10 \times 4 = 20$ — M1

60 + 20 = 80 cm² — A1

Tip: split the shape, find each area, then add.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

$\frac{1}{2} \times (8 + 14) \times 5$ — M1

= 55 cm² — A1

Tip: use the height between the parallel sides.

Question 4

$\frac{1}{2} \times (12 + 18) \times 8 = 120$ — M1

$120 \div 25 = 4.8$ — M1

5 boxes — A1

Tip: round up, because 4 boxes is not enough.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

$9 \times 4 = 36$ — M1

$\frac{1}{2} \times 6 \times x = 36$ — M1

$x = 12$ — A1

Tip: work out the known area first.

Question 6

$\frac{1}{2}(x + x + 6) \times 5 = 65$ — M1

$2x + 6 = 26$ — M1

$2x = 20$ — M1

$x = 10$ — A1

Tip: write the equation, then solve it.